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- Hierarchical models
  - Rats example
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- Exchangeability
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# Bayesian Statistics and Data Analysis

## Lecture 7

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Thanks to Aki Vehtari, Aalto University



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## Section 1

# Hierarchical models



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# Stan tutorial for hierarchical models

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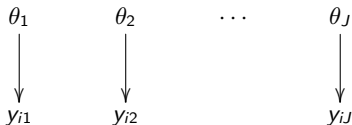
- Before Assignment 7, watch [Hierarchical Modelling in Stan: Predicting the Premier League](#)
- While watching, connect the example to:
  - partial pooling
  - group-level parameters
  - comparison with non-hierarchical models
  - Stan diagnostics for hierarchical models



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# Survival probabilities across hospitals

- Example: comparing hospitals
  - Hospital  $j$ : survival probability  $\theta_j$
  - Patient  $i$ 's survival outcome:  $y_{ij}$





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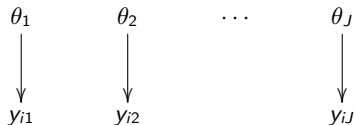
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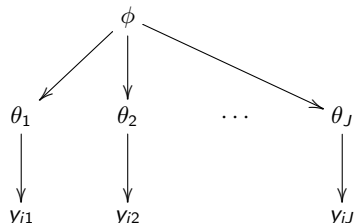
# Survival probabilities across hospitals

- Example: comparing hospitals

- Hospital  $j$ : survival probability  $\theta_j$
- Patient  $i$ 's survival outcome:  $y_{ij}$



- Model the  $\theta_j$  as similar, not identical



- A shared population distribution has hyperparameters  $\phi$
- Both  $\theta_j$  and  $\phi$  are unknown



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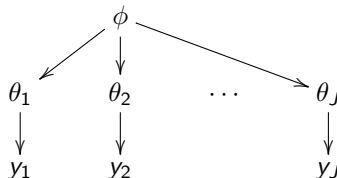
- $j = 1, \dots, J$ : groups;  $i = 1, \dots, n_j$ : observations in group  $j$
- $y_{ij}$ : individual observation;  $y_j$ : group data or a group summary;  $y = (y_1, \dots, y_J)$
- $\theta_j$ : group-specific parameter;  $\theta = (\theta_1, \dots, \theta_J)$
- $\phi$ : population hyperparameters

## For normal hierarchies:

- $\mu$ : population mean;  $\tau$ : between-group SD
- $\sigma$ : within-group SD of individual observations



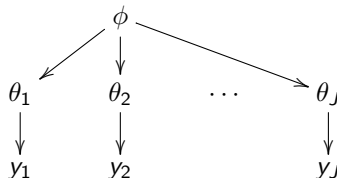
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- Given  $\phi$ , the  $\theta_j$  are independent with a common population distribution  $p(\theta_j | \phi)$



# Hierarchical model: assumptions



- Given  $\phi$ , the  $\theta_j$  are independent with a common population distribution  $p(\theta_j | \phi)$
- Given  $\theta$ , groups are independent, and  $y_j$  depends only on its own parameter  $\theta_j$
- The hyperprior  $p(\phi)$  describes uncertainty about the population distribution



Bayes' rule and the conditional-independence assumptions give

$$\begin{aligned} p(\theta, \phi \mid y) &\propto p(y \mid \theta) p(\theta \mid \phi) p(\phi) \\ &\propto p(\phi) \prod_{j=1}^J [p(\theta_j \mid \phi) p(y_j \mid \theta_j)] . \end{aligned}$$

- One hyperprior; each group contributes a population factor and a likelihood

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Bayes' rule and the conditional-independence assumptions give

$$\begin{aligned} p(\theta, \phi \mid y) &\propto p(y \mid \theta) p(\theta \mid \phi) p(\phi) \\ &\propto p(\phi) \prod_{j=1}^J [p(\theta_j \mid \phi) p(y_j \mid \theta_j)] . \end{aligned}$$

- One hyperprior; each group contributes a population factor and a likelihood
- Given  $\phi$  and  $y$ , the group parameters are independent
- Averaging over the shared, uncertain  $\phi$  generally makes the  $\theta_j$  dependent in the posterior
- Each group's data inform  $\phi$ , so other groups can inform its estimate of  $\theta_j$

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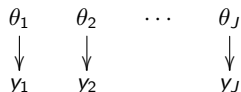
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# No, complete, and partial pooling

**No pooling** (separate)  
Independent priors; each  $\theta_j$  uses  
only group  $j$ 's data.





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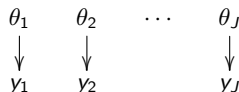
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# No, complete, and partial pooling

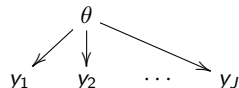
## No pooling (separate)

Independent priors; each  $\theta_j$  uses only group  $j$ 's data.



## Complete pooling (pooled)

All groups inform one common  $\theta$ .





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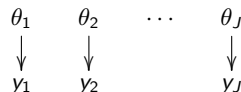
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# No, complete, and partial pooling

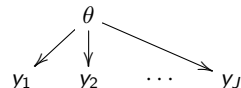
## No pooling (separate)

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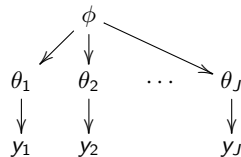
## Complete pooling (pooled)

All groups inform one common  $\theta$ .



## Partial pooling (hierarchical)

Distinct  $\theta_j$  share information through  $\phi$ .





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## Pair discussion: group size and pooling

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### Discuss in pairs (1–2 minutes).

Two groups have the same observed success rate:

$$\text{Group A: } \frac{1}{5} = 20\%, \quad \text{Group B: } \frac{100}{500} = 20\%.$$

Both share the same population information, with a population mean of 32%.

**Which estimate should move more toward the population mean? Why?**



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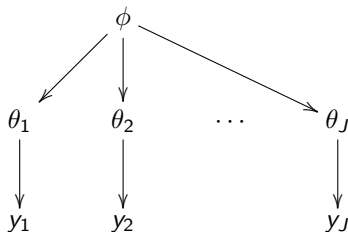
**Answer:** Group A. Its five observations provide weaker evidence, so the population information receives more weight.



# Predictive distributions for hierarchical models

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- Two types of predictive distributions
  1. A new observation in **an existing group**
  2. A new observation in **a new group**





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# Hierarchical binomial model: rats

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- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
  - familiar binomial model example



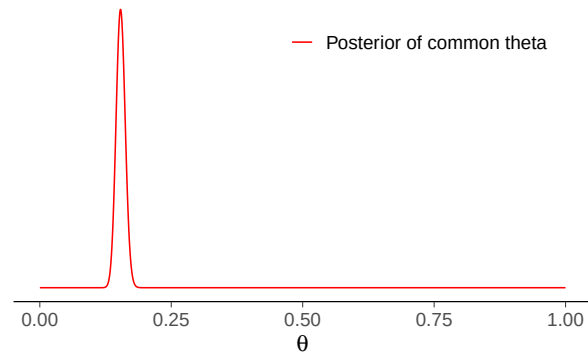
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- Medicine testing
- Type F344 female rats in control group given placebo
  - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
  - familiar binomial model example
- Experiment has been repeated 71 times

|      |       |      |       |      |      |       |       |       |      |
|------|-------|------|-------|------|------|-------|-------|-------|------|
| 0/20 | 0/20  | 0/20 | 0/20  | 0/20 | 0/20 | 0/20  | 0/19  | 0/19  | 0/19 |
| 0/19 | 0/18  | 0/18 | 0/17  | 1/20 | 1/20 | 1/20  | 1/20  | 1/19  | 1/19 |
| 1/18 | 1/18  | 2/25 | 2/24  | 2/23 | 2/20 | 2/20  | 2/20  | 2/20  | 2/20 |
| 2/20 | 1/10  | 5/49 | 2/19  | 5/46 | 3/27 | 2/17  | 7/49  | 7/47  | 3/20 |
| 3/20 | 2/13  | 9/48 | 10/50 | 4/20 | 4/20 | 4/20  | 4/20  | 4/20  | 4/20 |
| 4/20 | 10/48 | 4/19 | 4/19  | 4/19 | 5/22 | 11/46 | 12/49 | 5/20  | 5/20 |
| 6/23 | 5/19  | 6/22 | 6/20  | 6/20 | 6/20 | 16/52 | 15/46 | 15/47 | 9/24 |
| 4/14 |       |      |       |      |      |       |       |       |      |



## Pooled model

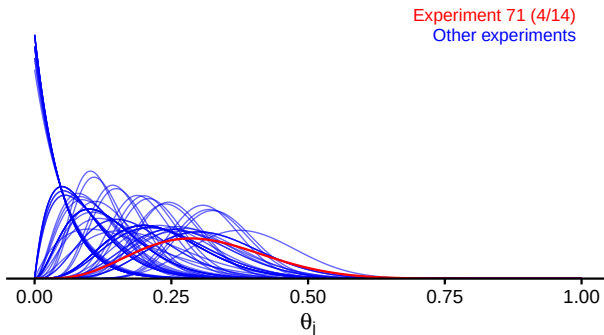


Complete pooling assumes one common probability for all experiments.



## Separate model

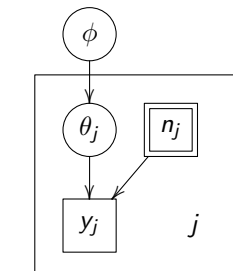
Existing-group posteriors



No pooling estimates each probability separately, without learning a shared population distribution.



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- Circle: unknown quantity
- Single box: observed quantity or a supplied estimate
- Double box: fixed, given quantity
- Arrow: the child's distribution depends on the parent
- Outer box (**plate**): repeat its contents for  $j = 1, \dots, J$

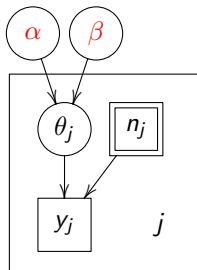
These are the conventions used in this lecture. A nested  $i$  plate repeats observations within each group.



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## Hierarchical binomial model: rats

- Here  $\phi = (\alpha, \beta)$ ;  $y_j$  counts affected rats out of known  $n_j$



$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$

$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$



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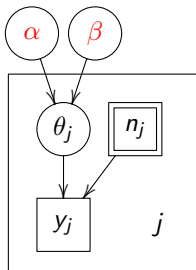
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## Hierarchical binomial model: rats

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$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$

$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$

$$p(\theta, \alpha, \beta \mid y) = \left[ \prod_{j=1}^J p(\theta_j \mid \alpha, \beta, y_j) \right] p(\alpha, \beta \mid y)$$

- One hyperposterior, not one per group
- The  $\theta_j$  are independent given  $\alpha, \beta, y$ , but generally dependent after averaging over  $\alpha, \beta$



For fixed  $\alpha, \beta > 0$ , the binomial likelihood gives

$$\begin{aligned} p(\theta_j \mid \alpha, \beta, y_j) &\propto \theta_j^{y_j} (1 - \theta_j)^{n_j - y_j} \theta_j^{\alpha - 1} (1 - \theta_j)^{\beta - 1} \\ &\propto \theta_j^{\alpha + y_j - 1} (1 - \theta_j)^{\beta + n_j - y_j - 1}. \end{aligned}$$

Thus

$$\theta_j \mid \alpha, \beta, y_j \sim \text{Beta}(\alpha + y_j, \beta + n_j - y_j).$$

- The population distribution supplies  $\alpha, \beta$
- Group  $j$  supplies  $y_j$  successes and  $n_j - y_j$  failures

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# Beta-binomial shrinkage

Write the population mean and concentration as

$$m = \frac{\alpha}{\alpha + \beta}, \quad \kappa = \alpha + \beta.$$

For  $n_j > 0$ ,

$$\begin{aligned} E(\theta_j \mid \alpha, \beta, y_j) &= \frac{\alpha + y_j}{\alpha + \beta + n_j} \\ &= \underbrace{\frac{n_j}{n_j + \kappa}}_{\text{data weight}} \frac{y_j}{n_j} + \underbrace{\frac{\kappa}{n_j + \kappa}}_{\text{population weight}} m. \end{aligned}$$

- The conditional posterior mean lies between  $y_j/n_j$  and  $m$
- With  $\alpha, \beta$  fixed, smaller groups receive more population weight
- Full Bayes also averages over uncertainty in  $\alpha, \beta$



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- Population prior  $\text{Beta}(\theta_j | \alpha, \beta)$
- Hyperprior  $p(\alpha, \beta)$ ?
  - $\alpha, \beta$  both affect the location and scale
  - BDA3 (pp. 110–111) uses, for  $\alpha, \beta > 0$ ,  
 $p(\alpha, \beta) \propto (\alpha + \beta)^{-5/2}$
  - This hyperprior is **improper**: its integral is infinite
  - BDA shows that the posterior is proper for these rat data; posterior propriety must be checked when using an improper prior
  - An example-specific choice, not a general default



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# Hierarchical inference with posterior draws

For  $s = 1, \dots, S$ :

1. Draw  $(\alpha^{(s)}, \beta^{(s)}) \sim p(\alpha, \beta \mid y)$
2. For every group  $j$ , draw independently conditional on this pair:

$$\theta_j^{(s)} \sim \text{Beta}(\alpha^{(s)} + y_j, \beta^{(s)} + n_j - y_j)$$

*Stan note:* If  $\theta_j$  is a model parameter, Stan returns its posterior draws directly.

**Share one hyperparameter draw across all groups at each step.**

$$\mathbb{E}[h(\theta_j) \mid y] = \int h(\theta_j) p(\theta_j \mid y) d\theta_j \approx \frac{1}{S} \sum_{s=1}^S h(\theta_j^{(s)}).$$

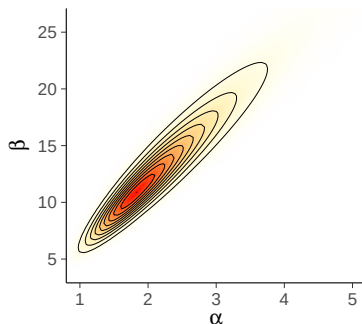
Choosing  $h(\theta) = \theta$  estimates a mean;  $h(\theta) = \mathbf{1}(\theta > c)$  estimates a probability.



# Rats: learning the population distribution

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The marginal of  $\alpha$  and  $\beta$

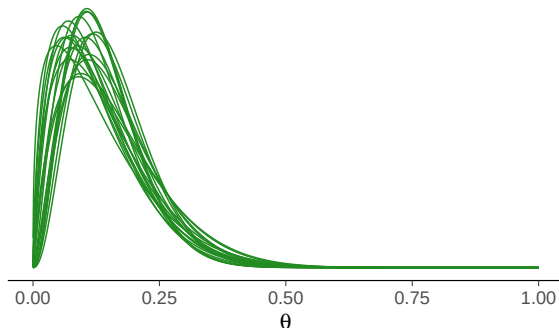


The hyperposterior  $p(\alpha, \beta \mid y)$  uses all 71 experiments.



# Rats: learning the population distribution

Beta( $\alpha, \beta$ ) given posterior draws of  $\alpha$  and  $\beta$

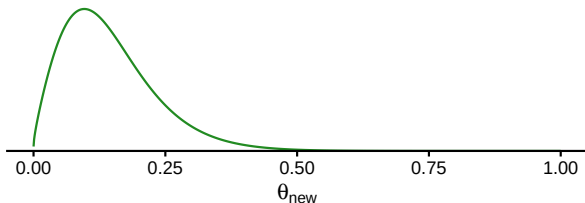


Each curve is Beta( $\alpha, \beta$ ) for one posterior draw of  $(\alpha, \beta)$ .



## New group's probability

Posterior predictive  $p(\theta_{\text{new}} | y)$



$$p(\theta_{\text{new}} | y) = \int_0^\infty \int_0^\infty p(\theta_{\text{new}} | \alpha, \beta) p(\alpha, \beta | y) d\alpha d\beta.$$

This averages the population curves over the hyperposterior. It predicts a new experiment's probability, before observing its data.

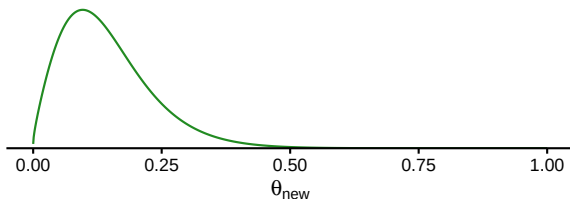


# Pair discussion: a new rat experiment

Discuss in pairs (1–2 minutes).

New group's probability

Posterior predictive  $p(\theta_{\text{new}} | y)$



A new experiment will include  $n_{\text{new}}$  rats.

**How would you use this distribution to predict the number of affected rats,  $y_{\text{new}}$ ? Describe the simulation steps.**

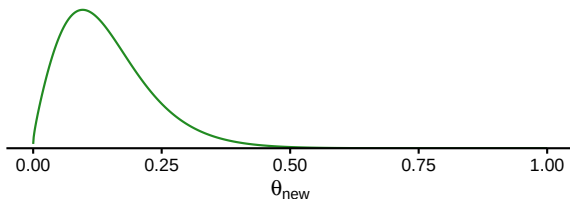


# Pair discussion: a new rat experiment

Discuss in pairs (1–2 minutes).

New group's probability

Posterior predictive  $p(\theta_{\text{new}} | y)$



**Answer:** Draw a probability from the green distribution, then draw a count:

$$\theta_{\text{new}}^{(s)} \sim p(\theta_{\text{new}} | y), \quad y_{\text{new}}^{(s)} \sim \text{Binomial}(n_{\text{new}}, \theta_{\text{new}}^{(s)}).$$

Repeat and summarize the simulated counts.



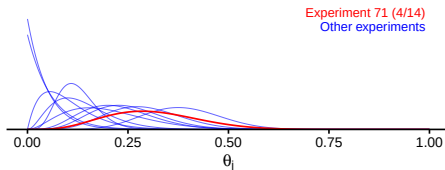
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# Rats: partial pooling

Experiment 71 (4/14), shown in red.

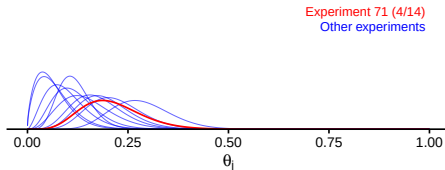
## Separate model

Existing-group posteriors



## Hierarchical model

Existing-group posteriors



Experiment 71 moves toward the population.  
Both plots use the same experiments and scales.

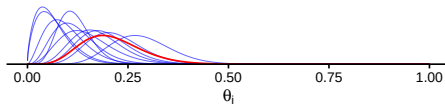


# Rats: existing and new experiments

## Hierarchical model

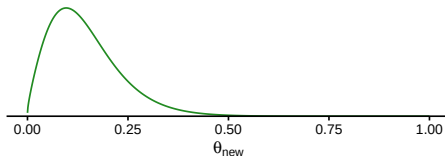
Existing-group posteriors

Experiment 71 (4/14)  
Other experiments



## New group's probability

Posterior predictive  $p(\theta_{\text{new}} | y)$



Existing  $p(\theta_j | y)$  uses group  $j$ 's data as well as the population.  
New  $p(\theta_{\text{new}} | y)$  has no new-group data.



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# Pair discussion: predictions for rats

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**Discuss in pairs (1–2 minutes).**

**What predictive distributions can we have?**

Think about both an existing experiment and a new similar experiment.

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**Discuss in pairs (1–2 minutes).**

## What predictive distributions can we have?

Think about both an existing experiment and a new similar experiment.

- Future outcomes or counts in an existing experiment, using  $p(\theta_j | y)$
- A new experiment's probability,  $p(\theta_{\text{new}} | y)$
- Outcomes or counts in that new experiment, also including binomial sampling variation



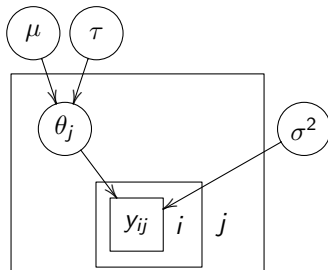
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# Hierarchical normal model: factory

- We evaluate the quality produced by six machines in a factory
- Assume hierarchical model
  - each machine has its own (average) quality  $\theta_j$  and **common variance**  $\sigma^2$

$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$

$$y_{ij} | \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2)$$



- Can be used to predict the future quality produced by each machine and quality produced by a new similar machine



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## Factory model: a complete hierarchy

One illustrative choice of **proper, independent priors** is

$$\begin{aligned}\mu &\sim \mathcal{N}(m_0, s_0^2), \\ \tau &\sim \text{HalfNormal}(0, A_\tau), \quad \sigma \sim \text{HalfNormal}(0, A_\sigma).\end{aligned}$$

The common-variance factory model then has

$$\begin{aligned}\theta_j &| \mu, \tau \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2), & j = 1, \dots, J, \\ y_{ij} &| \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), & i = 1, \dots, n_j.\end{aligned}$$

- $\text{HalfNormal}(0, A)$ : a  $\mathcal{N}(0, A^2)$  restricted to  $\mathbb{R}^+ = (0, \infty)$
- Observations are independent given the group means and  $\sigma$
- Fix  $m_0, s_0, A_\tau, A_\sigma$  using subject knowledge (e.g. prior predictive distribution)



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## A half-normal prior for a standard deviation

A simple way to restrict  $\tau$  to positive values is

$$Z \sim \mathcal{N}(0, A_\tau^2), \quad \tau = |Z|.$$

This defines  $\tau \sim \text{HalfNormal}(0, A_\tau)$  on  $\mathbb{R}^+ = (0, \infty)$ , with

$$\Pr(\tau < 1.96A_\tau) \approx 0.95.$$

- The 0 denotes the mean of the underlying normal  $Z$ , not of  $\tau$
- $A_\tau$  has the same units as the outcome
- Example: to put 95% prior probability below a between-machine SD of 20 quality units, choose  $A_\tau \approx 20/1.96 = 10.2$
- Choose  $A_\sigma$  separately for variation within a machine
- These scales are illustrative, not universal defaults



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## The prior on variation affects pooling

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- Small  $\tau$ : machine means lie close to  $\mu$ , encouraging stronger pooling
- Large  $\tau$ : machine means can differ more, allowing less pooling
- With few groups, the data may give little information about  $\tau$ , so its prior can matter substantially
- Check prior predictive outcomes and compare plausible prior scales
- A uniform prior on  $\tau$ , on  $\tau^2$ , or on  $\log \tau$  gives different assumptions; “flat” does not mean neutral

BDA3, Section 5.7 discusses priors for hierarchical variation. The half-normal specification here is one modelling choice.



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## Factory: observations and means

- In the common-variance model, observations are independent given  $\theta_j$  and  $\sigma$ :

$$y_{ij} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), \quad i = 1, \dots, n_j.$$

- Machine  $j$ 's sample mean and its sampling variance are

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij},$$

$$\text{Var}(\bar{y}_{.j} \mid \theta_j, \sigma) = \frac{\sigma^2}{n_j} = s_j^2.$$

SD of observations:  $\sigma$ ; SE of the mean:  $s_j = \sigma / \sqrt{n_j}$ .



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SD of observations:  $\sigma$ ; SE of the mean:  $s_j = \sigma / \sqrt{n_j}$ .

- Thus  $\bar{y}_{.j} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, s_j^2)$ . We condition on  $\sigma$  here; full Bayes also learns  $\sigma$  from the data.



## Normal shrinkage: the weights

For fixed  $\mu$ ,  $\tau > 0$  and  $\sigma > 0$ , let  $s_j = \sigma / \sqrt{n_j}$ .

$$E(\theta_j \mid \mu, \tau, \sigma, y) = w_j \bar{y}_{.j} + (1 - w_j) \mu, \quad w_j = \frac{\tau^2}{\tau^2 + s_j^2}.$$

- A precise machine mean (small  $s_j$ ) receives more weight
- Small between-machine variation  $\tau$  gives more weight to  $\mu$
- As  $\tau \rightarrow 0$ , the conditional mean approaches  $\mu$ ; as  $\tau \rightarrow \infty$ , it approaches  $\bar{y}_{.j}$
- In full Bayes, average over  $p(\mu, \tau, \sigma \mid y)$
- Strong pooling makes the machine means similar; their common value can still be uncertain because  $\mu$  is unknown

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## Normal shrinkage: a numerical example

Suppose machine 1 has  $\bar{y}_{.1} = 28$  from  $n_1 = 4$  observations. For illustration, **fix**  $\mu = 8$ ,  $\tau = 5$  and  $\sigma = 30$ , so  $s_1 = 30/\sqrt{4} = 15$ .

$$w_1 = \frac{5^2}{5^2 + 15^2} = \frac{25}{250} = 0.10,$$

$$m_1 = 0.10 \cdot 28 + 0.90 \cdot 8 = 10,$$

$$v_1 = (s_1^{-2} + \tau^{-2})^{-1} = \left( \frac{1}{225} + \frac{1}{25} \right)^{-1} = 22.5.$$

Posterior precision = data precision + prior precision.

The conditional posterior is  $\theta_1 \mid \mu, \tau, \sigma, y \sim \mathcal{N}(m_1, v_1)$ : mean 10, SD  $\sqrt{22.5} \approx 4.74$ .

- The noisy machine mean receives 10% of the weight
- The full factory posterior also averages over  $\mu, \tau, \sigma$



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## Pair discussion: precision and shrinkage

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### Discuss in pairs (1–2 minutes).

Two machines have the same sample mean, different from  $\mu$ . They share a within-machine SD  $\sigma$ , but different sample sizes give different standard errors of their means:

$$s_1 = 5, \quad s_2 = 20.$$

Hold  $\mu$ ,  $\tau > 0$  and  $\sigma > 0$  fixed.

**Which machine's estimate is shrunk more toward  $\mu$ ?**  
**Why?**



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## Pair discussion: precision and shrinkage

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$$s_1 = 5, \quad s_2 = 20.$$

Hold  $\mu$ ,  $\tau > 0$  and  $\sigma > 0$  fixed.

**Which machine's estimate is shrunk more toward  $\mu$ ? Why?**

**Answer:** Machine 2. Its larger standard error means less precise evidence, so its sample mean receives less weight and  $\mu$  receives more weight.



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## Prediction for an existing machine

In the common-variance model, distinguish the mean  $\theta_j$  from a future observation  $\tilde{y}_j$ :

$$p(\tilde{y}_j | y) = \int \int p(\tilde{y}_j | \theta_j, \sigma) p(\theta_j, \sigma | y) d\theta_j d\sigma.$$

For each posterior draw  $s$ :

1. Take  $(\theta_j^{(s)}, \sigma^{(s)})$  from the **same joint posterior draw**
2. Draw  $\tilde{y}_j^{(s)} \sim \mathcal{N}(\theta_j^{(s)}, (\sigma^{(s)})^2)$

- The  $\theta_j^{(s)}$  summarize uncertainty about the machine's mean
- The  $\tilde{y}_j^{(s)}$  also include variation in a new observation



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## Prediction for a new similar machine

With no observations yet from the new machine,

$$p(\theta_{\text{new}} \mid y) = \int \int p(\theta_{\text{new}} \mid \mu, \tau) p(\mu, \tau \mid y) d\mu d\tau,$$

$$p(\tilde{y}_{\text{new}} \mid y) = \int \int p(\tilde{y}_{\text{new}} \mid \theta_{\text{new}}, \sigma) p(\theta_{\text{new}}, \sigma \mid y) d\theta_{\text{new}} d\sigma.$$

For each posterior draw  $s$ :

1. Take  $(\mu^{(s)}, \tau^{(s)}, \sigma^{(s)})$  jointly from  $p(\mu, \tau, \sigma \mid y)$
2. Draw the new mean:  $\theta_{\text{new}}^{(s)} \sim \mathcal{N}(\mu^{(s)}, (\tau^{(s)})^2)$
3. For a new observation, also draw  $\tilde{y}_{\text{new}}^{(s)} \sim \mathcal{N}(\theta_{\text{new}}^{(s)}, (\sigma^{(s)})^2)$

Predicting a new mean stops at step 2; predicting an observation includes step 3.



# Pair discussion: a new machine

---

## Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality,  $\theta_7$
- The quality of its next item,  $\tilde{y}_7$

**How do the simulation steps differ? What additional uncertainty enters the second prediction?**

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## Pair discussion: a new machine

---

### Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality,  $\theta_7$
- The quality of its next item,  $\tilde{y}_7$

**How do the simulation steps differ? What additional uncertainty enters the second prediction?**

**Answer:** Both start by drawing a new machine mean using posterior draws of the population parameters. Predicting an item also adds within-machine variation:

$$\tilde{y}_7^{(s)} \sim \mathcal{N}(\theta_7^{(s)}, (\sigma^{(s)})^2).$$



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## Section 2

# Exchangeability



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# Independence and exchangeability

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1. **Independence:** knowing one outcome does not change the distribution of another.
2. **Exchangeability:** swapping the labels or positions of observations leaves their joint distribution unchanged.



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- Motivates a **symmetric joint model**
  - for observations whose labels carry no relevant information
  - for group parameters that we treat symmetrically a priori
- Less restrictive than **IID** (independent and identically distributed)
- IID implies exchangeability, but exchangeability need not imply independence
- Independence alone is not enough: the distributions may differ



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# Exchangeability

- *Exchangeability*

Random variables  $\theta_1, \dots, \theta_J$  (or  $y_1, \dots, y_J$ ) are **exchangeable** if the joint distribution  $p$  is invariant to the permutation of indices

i.e

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_{\pi(1)}, \theta_{\pi(2)}, \theta_{\pi(3)})$$

for any permutation  $\pi$  of the indices. E.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$



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$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

Can we come up with a situation where this doesn't hold?



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**Can we come up with a situation where this doesn't hold?**  
E.g. when we have a trend over indices (like in time series data)



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for any permutation  $\pi$  of the indices. E.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

**Can we come up with a situation where this doesn't hold?**

E.g. when we have a trend over indices (like in time series data)

- Exchangeability implies symmetry:  
If there is no information which can be used *a priori* to separate  $\theta_j$  from each other, we can assume exchangeability. ("Ignorance implies exchangeability")



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## Exchangeability: de Finetti's theorem

- **Bernoulli case:** let  $(X_i)_{i=1}^{\infty}$  be an infinite exchangeable sequence with  $X_i \in \{0, 1\}$
- There is a random  $\theta \in [0, 1]$  such that the  $X_i$  are **conditionally IID**:

$$X_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

- If  $\theta$  has a density  $p(\theta)$ , then for every finite  $J$ ,

$$p(x_1, \dots, x_J) = \int_0^1 \left[ \prod_{i=1}^J \theta^{x_i} (1 - \theta)^{1-x_i} \right] p(\theta) d\theta$$

- The theorem does not choose the mixing distribution: a Beta prior is a modelling choice, not a consequence



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## De Finetti's takeaway

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**Exchangeability provides a justification for modelling observations as independent conditional on a shared unknown parameter, even though they need not be independent marginally.**

1. **Usually, we start with a Bayesian model:** an unknown coin bias  $\theta$ , a prior for  $\theta$ , and independent flips given  $\theta$ .
2. **De Finetti starts with symmetry:** the order of the flips carries no information.
3. For an **infinite exchangeable Bernoulli sequence**, the theorem says its distribution must admit exactly that Bayesian representation: a distribution over  $\theta$  and IID flips conditional on  $\theta$ .



# Pair discussion: learning about the coin

---

## Discuss in pairs (1–2 minutes).

The coin's bias has prior  $\theta \sim \text{Beta}(1, 1)$ , and the flips are independent given  $\theta$ .

**Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?**

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# Pair discussion: learning about the coin

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The coin's bias has prior  $\theta \sim \text{Beta}(1, 1)$ , and the flips are independent given  $\theta$ .

**Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?**

**Answer:** We learn about the shared unknown bias. After observing heads,  $\theta \mid \text{heads} \sim \text{Beta}(2, 1)$ , changing the predictive probability from  $1/2$  to  $2/3$ .

The coin has not changed. Independence holds when  $\theta$  is fixed; averaging over an uncertain  $\theta$  creates dependence.

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# Coin flips: exchangeable but dependent

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Let  $\theta \sim \text{Beta}(1, 1)$  and, given  $\theta$ , let the flips be IID Bernoulli( $\theta$ ). Write  $H_i$  for heads on flip  $i$ .

$$\Pr(H_1) = \mathbb{E}(\theta) = \frac{1}{2}.$$

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$$\Pr(H_1) = \mathbb{E}(\theta) = \frac{1}{2}.$$

After observing  $H_1$ ,

$$\theta \mid H_1 \sim \text{Beta}(2, 1), \quad \Pr(H_2 \mid H_1) = \mathbb{E}(\theta \mid H_1) = \frac{2}{3}.$$

- Marginally  $\Pr(H_2) = 1/2$ , so the flips are dependent
- Their joint distribution remains unchanged when we swap the flips
- Observing heads updates the shared unknown bias; one toss does not physically affect the next



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# Exchangeability

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- Exchangeability does not require identical experiment probabilities
- Suppose experiments came from two laboratories with different conditions, but their laboratory memberships are unknown
- If we treat these unknown memberships symmetrically across experiments, we can model the probabilities  $\theta_j$  as **exchangeable a priori**
- A model with unknown laboratory membership can represent clustering among experiments from the same laboratory
- This concerns  $\theta_j$ , not the counts  $y_j$ : different sample sizes  $n_j$  can make the counts non-exchangeable



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# Exchangeability and additional information

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- Bioassay: doses  $x_j$  and group sizes  $n_j$  are fixed design information;  $y_j$  counts deaths

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## Exchangeability and additional information

- Bioassay: doses  $x_j$  and group sizes  $n_j$  are fixed design information;  $y_j$  counts deaths
- Given  $\alpha, \beta$ , the counts are independent, but generally not identically distributed:

$$y_j \mid \alpha, \beta, n_j, x_j \sim \text{Binomial}(n_j, \pi_j), \quad j = 1, \dots, J$$

where  $\text{logit}(\pi_j) = \alpha + \beta x_j$ .

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^J p(y_j \mid \alpha, \beta, n_j, x_j)$$



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$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^J p(y_j \mid \alpha, \beta, n_j, x_j)$$

- Reordering complete records  $(x_j, n_j, y_j)$  leaves the likelihood unchanged; swapping only counts between doses generally changes it



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# Hierarchical exchangeability

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- Rat example: experiments  $j = 1, \dots, J$ , each with  $n_j$  rats



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# Hierarchical exchangeability

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  - Within experiment  $j$ , outcomes are IID Bernoulli( $\theta_j$ ) given  $\theta_j$ ; they need not be exchangeable across experiments



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  - Within experiment  $j$ , outcomes are IID Bernoulli( $\theta_j$ ) given  $\theta_j$ ; they need not be exchangeable across experiments
  - The group probabilities  $\theta_j$  are modelled exchangeably a priori



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- Rat example: experiments  $j = 1, \dots, J$ , each with  $n_j$  rats
- Within experiment  $j$ , outcomes are IID Bernoulli( $\theta_j$ ) given  $\theta_j$ ; they need not be exchangeable across experiments
- The group probabilities  $\theta_j$  are modelled exchangeably a priori
- Counts  $y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$  need not be exchangeable when  $n_j$  differs
- This assumes comparable experiments; grouping alone does not justify a shared population distribution



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- **Conditional exchangeability**

- Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
- For fixed covariates, use a conditional model  $p(y \mid x)$ ; adding  $x_i$  does not make  $(x_i, y_i)$  exchangeable



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- Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
- For fixed covariates, use a conditional model  $p(y \mid x)$ ; adding  $x_i$  does not make  $(x_i, y_i)$  exchangeable

- **Partial exchangeability**

- Symmetry within specified groups, not under arbitrary exchanges across groups
- The models here additionally assume exchangeable group parameters, possibly given group-level covariates



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## Section 3

# Computational aspects



- Consider the **prior hierarchy**, before observing data:

$$\tau \sim p(\tau), \quad \tau > 0, \quad \theta_j \mid \tau \sim \mathcal{N}(0, \tau) \quad (\text{SD form})$$

- Small  $\tau$ : group effects lie near 0; large  $\tau$ : group effects are more spread out
- If plausible  $\tau$  values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel

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$$\tau \sim p(\tau), \quad \tau > 0, \quad \theta_j \mid \tau \sim \mathcal{N}(0, \tau) \quad (\text{SD form})$$

- Small  $\tau$ : group effects lie near 0; large  $\tau$ : group effects are more spread out
- If plausible  $\tau$  values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel
- With weak information about group effects, the **posterior can retain** this funnel geometry
- Not every hierarchical posterior is funnel-shaped: the likelihood also matters

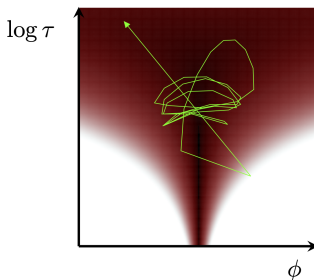
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## Why is a funnel problematic?

- Changing curvature makes it hard for HMC to explore both the narrow and wide regions efficiently
- Post-warmup divergences warn of potentially unreliable exploration and biased posterior estimates
- Good  $\hat{R}$  and ESS do not cancel divergence warnings



Betancourt (2020) [Stan diagnostics](#) [HMC demo](#)

In this figure,  $\phi$  denotes one group effect  $\theta_j$ .

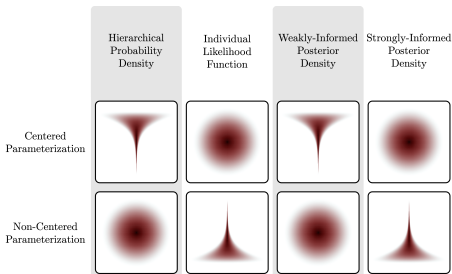


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## Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered**: often better with strong group information

$$\theta_j \sim N(\mu, \tau) \quad (\text{SD form})$$





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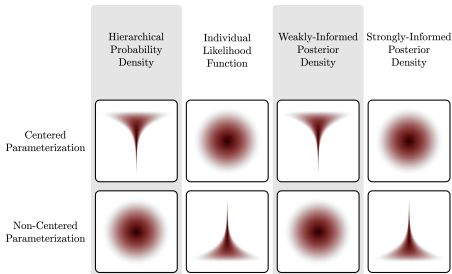
## Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered:** often better with strong group information

$$\theta_j \sim N(\mu, \tau) \quad (\text{SD form})$$

- **Non-centered:** often helps weakly informed group effects

$$\eta_j \sim N(0, 1), \quad \theta_j = \mu + \tau \eta_j$$





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## Why non-centering preserves the model

Start with  $\eta_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ , independent of the hyperparameters a priori, and set

$$\theta_j = \mu + \tau \eta_j, \quad \tau > 0.$$

Conditional on  $\mu, \tau$ , an affine transformation of a normal variable is normal, with

$$\begin{aligned} \mathbb{E}(\theta_j \mid \mu, \tau) &= \mu + \tau \mathbb{E}(\eta_j) = \mu, \\ \text{Var}(\theta_j \mid \mu, \tau) &= \tau^2 \text{Var}(\eta_j) = \tau^2. \end{aligned}$$

Thus the  $\theta_j$  are conditionally independent normals with mean  $\mu$  and SD  $\tau$ , as in the centered hierarchy.

Keep the same hyperpriors and likelihood, evaluated at  $\theta_j = \mu + \tau \eta_j$ . Only the sampling coordinates change.



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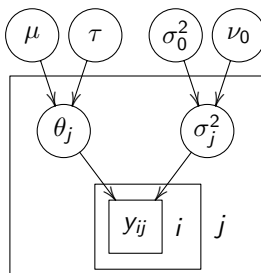
## Section 4

### Additional material



## Factory: separate variances

- Six machines with means  $\theta_j$  and **own variances**  $\sigma_j^2$



$$y_{ij} \mid \theta_j, \sigma_j^2 \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

$$\theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$

$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

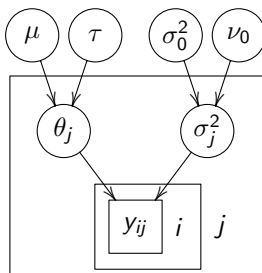
- Hierarchical models
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## Factory: separate variances

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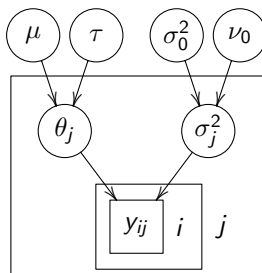
$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

- **Pair discussion (1–2 minutes):** What predictive distributions can we have?



## Factory: separate variances

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- **Pair discussion (1–2 minutes):** What predictive distributions can we have?
- Observations from existing or new similar machines



# Eight schools: historical SAT coaching study

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- Historical example from BDA3, Section 5.5
  - randomized coaching experiments in eight high schools
  - question: does short-term coaching improve verbal test scores?
  - estimate a coaching effect for each school



# Eight schools: historical SAT coaching study

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- Historical example from BDA3, Section 5.5
  - randomized coaching experiments in eight high schools
  - question: does short-term coaching improve verbal test scores?
  - estimate a coaching effect for each school
- SAT-V in this study (the historical verbal test)
  - standardized multiple-choice test
  - score range 200–800, with mean about 500 and SD about 100
  - pre-test PSAT scores in verbal (V) and mathematics (M) help adjust for students' initial performance



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- Effectiveness of the SAT coaching
  - students took pre-tests PSAT-M and PSAT-V
  - some students received coaching
  - a regression in each school adjusts for pre-test scores and estimates the coaching effect  $y_j$  and its standard error  $\sigma_j$
  - use a normal approximation for  $y_j$ , treating the estimated sampling variances  $\sigma_j^2$  as known
  - $y_j$  is an adjusted effect estimate, not a sample mean;  $\sigma_j$  is not the SD of individual students' scores



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- Data:

| School     | A  | B  | C  | D  | E  | F  | G  | H  |
|------------|----|----|----|----|----|----|----|----|
| $y_j$      | 28 | 8  | -3 | 7  | -1 | 1  | 18 | 12 |
| $\sigma_j$ | 15 | 10 | 16 | 11 | 9  | 11 | 10 | 18 |

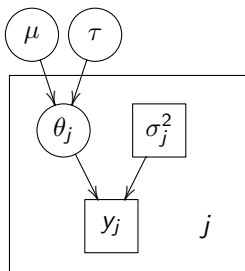


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# Hierarchical normal model for school effects

$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau)$$

$$y_j | \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$



The observed  $y_j$  is a school effect estimate;  $\sigma_j$  is its reported standard error, treated as known.

Here  $\mathcal{N}(\mu, \tau)$  is written in mean–SD form;  $\tau$  is the between-school SD.



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## Normal shrinkage: completing the square

For fixed  $\mu$  and  $\tau > 0$ , with known sampling SD  $\sigma_j > 0$ ,

$$y_j \mid \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2), \quad \theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2).$$

Here  $\mathcal{N}$  uses variance as its second argument.

$$\begin{aligned} p(\theta_j \mid \mu, \tau, y) &\propto \exp \left[ -\frac{1}{2} \left\{ \frac{(y_j - \theta_j)^2}{\sigma_j^2} + \frac{(\theta_j - \mu)^2}{\tau^2} \right\} \right] \\ &\propto \exp \left[ -\frac{(\theta_j - m_j)^2}{2v_j} \right], \end{aligned}$$

where completing the square gives

$$v_j = \left( \frac{1}{\sigma_j^2} + \frac{1}{\tau^2} \right)^{-1}, \quad m_j = v_j \left( \frac{y_j}{\sigma_j^2} + \frac{\mu}{\tau^2} \right).$$

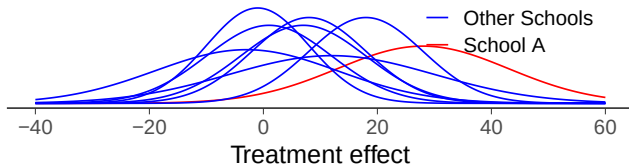
Thus  $\theta_j \mid \mu, \tau, y \sim \mathcal{N}(m_j, v_j)$ .



# Eight schools: pooling assumptions

How does the posterior for School A (red) change?

## Separate model



**No pooling:** each school's effect uses its own data. School A's estimate is high but uncertain.

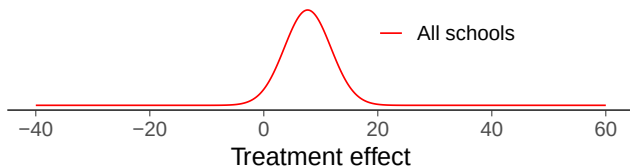
The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on  $\mu$  and  $\tau > 0$ , rather than the illustrative half-normal prior in the factory example.



# Eight schools: pooling assumptions

How does the posterior for School A (red) change?

## Pooled model



**Complete pooling:** all schools share a common effect. School A's posterior is identical to the others.

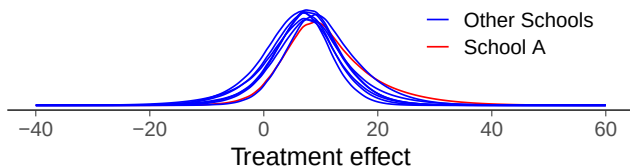
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# Eight schools: pooling assumptions

How does the posterior for School A (red) change?

## Hierarchical model



**Partial pooling:** the marginal posteriors  $p(\theta_j | y)$  share information while allowing different effects.

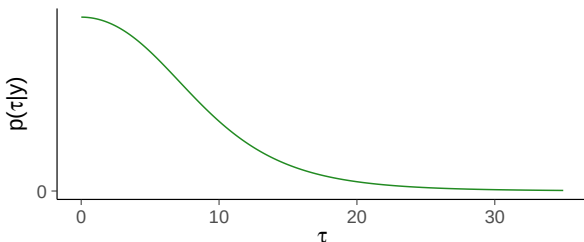
The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on  $\mu$  and  $\tau > 0$ , rather than the illustrative half-normal prior in the factory example.



## Eight schools: uncertainty about variation

Does the peak near zero establish that all effects are equal?

Marginal posterior  $p(\tau|y)$



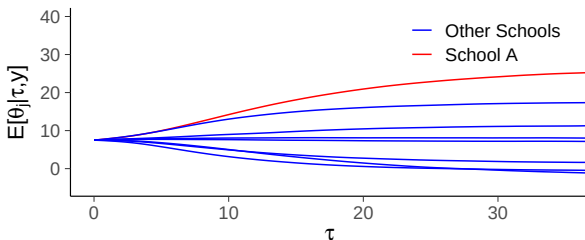
- This is  $p(\tau | y)$ , with  $\mu$  and the school effects integrated out
- A range of between-school SDs remains plausible
- Full Bayes averages over this distribution; fixing  $\tau$  at its mode would impose much stronger pooling



# Eight schools: conditional posterior means

What happens to the school estimates as  $\tau$  changes?

Conditional means  $E[\theta_j|\tau, y]$



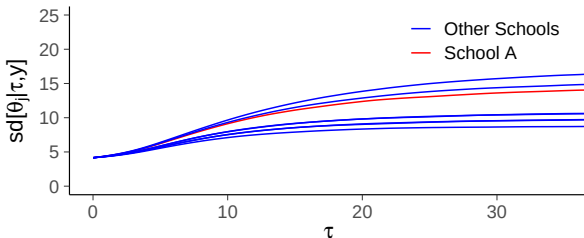
- Curves show  $E(\theta_j | \tau, y)$ :  $\tau$  is fixed and  $\mu$  is integrated out
- Small  $\tau$ : means come together near the pooled estimate, 7.7
- Large  $\tau$ : means approach the separate estimates  $y_j$



## Eight schools: uncertainty given $\tau$

Does small  $\tau$  make the common effect precisely known?

### Conditional standard deviations $sd[\theta_j|\tau,y]$



- Curves show  $sd(\theta_j \mid \tau, y)$ , with uncertainty in  $\mu$  included
- As  $\tau \rightarrow 0$ , effects coincide, but their common value is uncertain:

$$sd(\theta_j \mid \tau, y) \longrightarrow \left( \sum_{k=1}^J \frac{1}{\sigma_k^2} \right)^{-1/2} \approx 4.1.$$